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Lumbar Microdiscectomy and Postoperative Activity Restrictions

A Randomized Controlled Trial

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Study Design. Single-blinded randomised controlled trial.

Objective. To establish the effect of postoperative mobility restrictions on outcome after lumbar microdiscectomy by comparing sitting and activity restrictions to no restrictions for the first month after surgery.

Summary of Background Data. Lumbar microdiscectomy effectively treats lumbar radiculopathy, improving leg pain and functional outcomes. However, 20% of patients experience residual sciatica and 5% require redo discectomy. Persistent sciatica causes suffering, increases health care costs, and results in work absenteeism. While strategies to prevent reherniation include postoperative mobility restriction, evidence is limited regarding efficacy. Most surgeons still advise sitting or lifting restrictions after microdiscectomy.

Methods. Two hundred patients (ages 18–75) undergoing unilateral microdiscectomy were randomised 1:1 to restricted (n = 101) or unrestricted (n = 99) groups. Restricted patients received limitations on sitting (15–30 mins per two hours), lifting (< 5 kg), and strenuous activities for two weeks. Unrestricted patients resumed normal activities as tolerated. All patients wore activity monitors (ActiV8) for one month. Assessments at baseline, day 1, and 1, 3, 6, and 12 months included VAS pain scores, Oswestry Disability Index, and quality-of-life questionnaires. The primary outcome was a composite of reduced pain, functional improvement, and absence of further interventions at 12 months.

Results. At one year, the primary composite outcome showed no significant difference between groups [restricted (41.6%) *vs*. unrestricted (36.4%), $P = 0.45$]. Secondary outcomes for restricted *versus* unrestricted groups, respectively, including reherniation rates (10.1% *vs*. 14.1%, $P = 0.61$), pain measures (VAS back reduction to 23.5 pts *vs*. 24.5 pts, $P = 0.83$), functional improvements (SF-12 PCS 50.3 *vs*. 49.7 pts at one year, $P = 0.57$), and reoperation rates (2.9% *vs*. 5.5%, $P = 0.68$) were similar. Activity monitoring revealed poor adherence to restrictions (10%) with no significant differences in sitting duration or other activities between groups (4102 *vs*. 4140 mins/wk, $P = 0.89$).

Conclusions. Liberalizing postoperative restrictions following lumbar microdiscectomy does not compromise outcomes. These findings support patient-driven recovery guided by comfort rather than rigid restrictions, potentially standardizing care guidelines and facilitating faster return to activities without compromising safety.

Key Words: lumbar microdiscectomy, postoperative restrictions, randomized controlled trial, activity limitations, spine surgery, discectomy outcomes, early mobilization, lumbar radiculopathy, rehabilitation, VAS pain scores.

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Lumbar microdiscectomy is an effective treatment for lumbar radiculopathy, significantly improving leg pain, functional outcomes, and quality of life.[1] However, up to 20% of patients will have residual sciatica after surgery[2] and 5% will require redo discectomy for recurrent disc prolapse.[1–3] Persistent or recurrent sciatica leads to emotional and physical suffering for the patient, increases health care costs, and work absenteeism.[4]

Strategies to prevent reherniation include postoperative mobility restriction, particularly the avoidance of prolonged sitting,[5] however, evidence is limited as to its efficacy.[6] Retrospective studies have demonstrated these mobility restrictions may not be necessary,[7] however, the majority of surgeons still advise some form of sitting or lifting restrictions after microdiscectomy.[8]

To establish the effect of postoperative mobility restrictions on outcome after lumbar microdiscectomy, we performed a randomized trial comparing sitting and activity restrictions to no restrictions for the first month after surgery. Modern wearable accelerometers facilitated objective monitoring of adherence to activity restrictions.

MATERIALS AND METHODS

A randomized controlled, single blinded trial was undertaken across three centers in Melbourne, Australia to assess the effect of postoperative activity restrictions on outcome after lumbar microdiscectomy. Patients undergoing unilateral microdiscectomy for lumbar radiculopathy were randomized to receive instructions limiting their activity and sitting time (restricted group), or to no specific instructions (unrestricted group). We enrolled patients aged 18 to 75 years, with lumbar radicular pain with concordant MRI of lumbar disc herniation at L3/L4, L4/L5 or L5/S1, who subsequently underwent lumbar microdiscectomy. Patients with a previous history of lumbar spine surgery, spinal infection or spinal fracture were excluded.

Patients were randomized to the restricted or unrestricted group in a 1:1 ratio, with a sealed envelope given to patients immediately after surgery that gave instructions on their postoperative mobility. All patients underwent standard open unilateral lumbar microdiscectomy. Patients in the restricted group were given the following instructions to be followed for the first postoperative month:

1. Limit sitting to no more than 15 to 30 minutes in any two-hour period.
2. Avoid bending, lifting, twisting, pulling, or pushing objects weighing more than five kg.
3. Refrain from engaging in heavy household tasks such as vacuuming, doing laundry, or making beds.
4. Avoid any strenuous sexual activity for the first two weeks after surgery.

Patients in the unrestricted group were encouraged to resume normal activities, including sitting, exercising, returning to work, and other daily tasks, as soon as they felt ready.

A wearable electronic activity monitor (Activ8, 2 M Engineering, the Netherlands) was worn by patients in both groups for the first postoperative month. The device measured the amount of time sitting, standing, running, walking, and lying during the day. Adherence to restrictions was defined as sitting limited to no more than 30 minutes in any two-hour period.

Patients were assessed neurologically by a "blinded" clinician preoperatively, and at day 1, 1 month, 3 months, 6 months and 12 months postsurgery. As well, the following questionnaires were completed at the same time; visual analog scale (VAS) for leg pain (VAS leg and VAS other leg),[9] VAS for back pain (VAS back),[9] ODI (Oswestry Disability Index),[10] and SF-12[11] and EuroQol (EQ-5D)[12] quality of life questionnaires. The primary outcome was a composite of a reduction of 18 points in VAS back, a reduction of 25 points in VAS leg, a 15-point improvement in ODI, and no further intervention at the same level at one year postsurgery.

Secondary outcomes included the incidence of recurrent herniation requiring surgery, nerve root sleeve or epidural injection at the index level, within the first 12 months, VAS leg, VAS back, ODI, and time to return to work. A secondary "as treated" analysis was planned, classifying patients based on their observed adherence to the restriction instructions using the electronic activity monitor.

We calculated that 210 patients would be required to have 80% power to detect a 20% difference in primary outcome between groups (assuming an 80% chance of the primary outcome in the control group), accounting for an anticipated 30% dropout rate and an α of 0.05.

Statistical analyses were performed using R Studio v. 2024.12.1 (R Core Team, 2025) following intention-to-treat principles. Baseline characteristics were compared using the χ^2 test for categorical variables and the Mann-Whitney U tests for continuous variables based on data distribution. The composite primary outcome was analysed using the Fisher exact test, while secondary outcomes were calculated with paired t tests for within-group changes and independent t tests for between-group comparisons for continuous variables and the Fisher exact test and χ^2 tests for categorical variables. Sensitivity analyses for the primary composite outcome were calculated using the last observation carried forward method (LOCF), best-value and worst-value replacement scenarios for fixed value imputation.

Secondary per-protocol analysis classified patients as "adherent" or "nonadherent" to activity restrictions based on electronic activity monitoring regardless of randomization assignment. Between-group comparisons used t tests for continuous outcomes and χ^2 tests for categorical outcomes.

All patients provided informed written consent before randomization, and the study was performed under institutional ethics approval (15379L).

RESULTS

Between 2016 and 2022, 200 patients were randomized to either the restricted (n = 101) or unrestricted arm (n = 99) following lumbar microdiscectomy (Figure 1). Approximately 29% of the cohort was lost to follow-up by 12 months (attrition of 31 and 27 patients for restrictions *vs*. unrestricted cohorts).

Baseline characteristics were well-balanced between groups (Table 1). The mean age was 47 years, and there was no sex bias. The majority of procedures were performed at the L4/L5 (39.5%) and L5/S1 (49%) levels by consultant neurosurgeons (74.6%). Baseline VAS leg [restricted (72.2 ± 20.6) *vs*. unrestricted (73.8 ± 22.3)], VAS back [restricted (63.3 ± 25.4) *vs*. unrestricted (59.3 ± 26.9)],

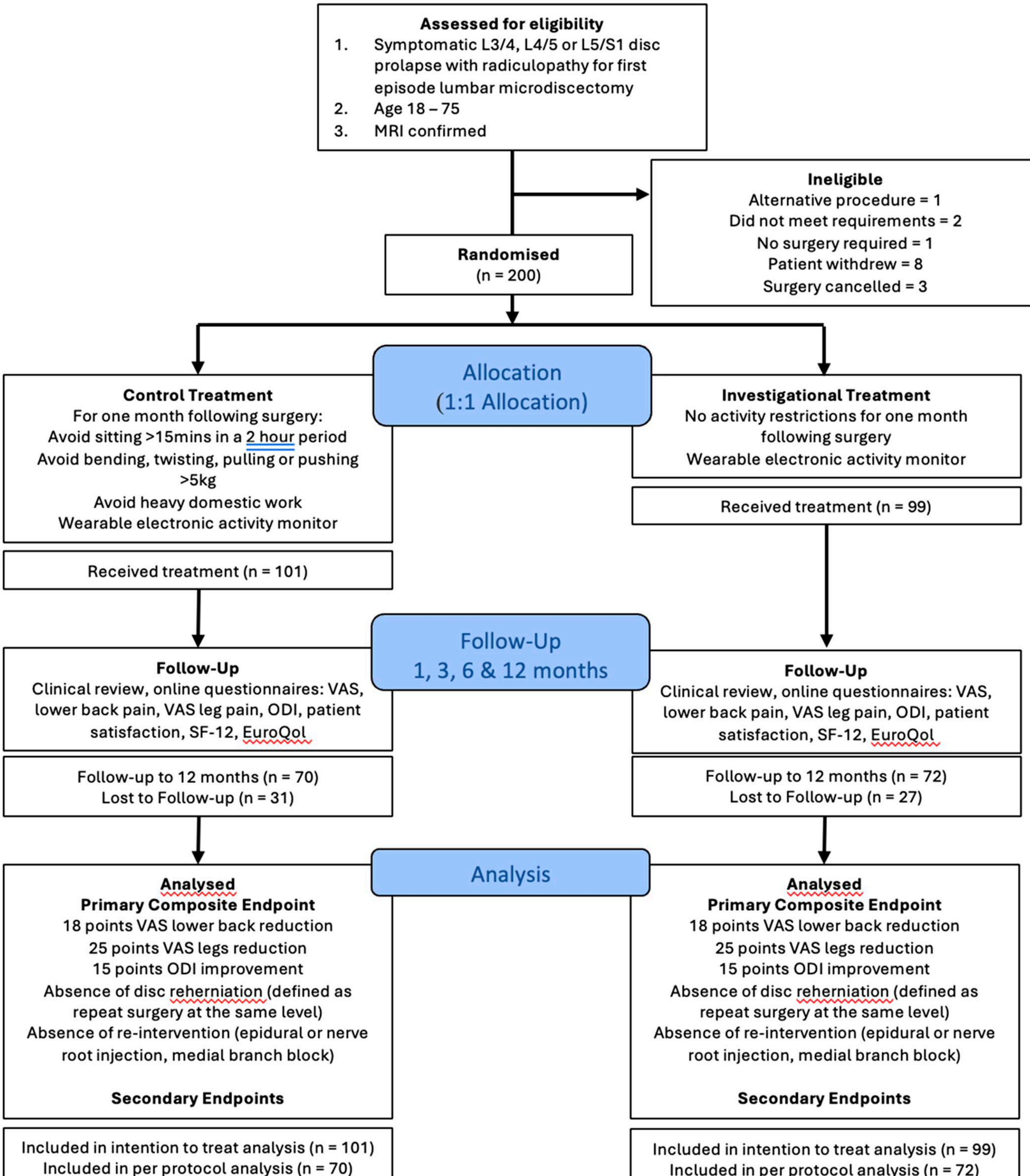

Figure 1. CONSORT flow diagram.

TABLE 1. Cohort Baseline and Operative Characteristics

Baseline Characteristics			
Comparison of Restrictions *Vs*. No Restrictions Groups			
Characteristic	**Overall (n = 200)**	**Restrictions (n = 101)**	**No Restrictions (n = 99)**
Age	47 ± 14.3 yr	46.3 ± 13.6 yr	48.5 ± 15.0 yr
Sex	94/99 (48.7% M, 51.3% F)	47/51 (47.9% M, 52% F)	48/49 (49.5% M, 50.5% F)
BMI (kg/m²)	29.0 ± 6.8	29.0 ± 7.5	29.0 ± 6.0
Smoker (%)	47 (23.5)	28 (27.7)	19 (19.1)
Baseline VAS back pain	61.3 ± 26.2	63.3 ± 25.4	59.3 ± 26.9
Baseline VAS leg pain	73.0 ± 21.4	72.2 ± 20.6	73.8 ± 22.3
Baseline ODI score (%)	48.6 ± 17.5	46.1 ± 17.2	51.1 ± 17.8
Baseline SF-12	—	50.7 ± 6.7 (PCS) 48.9 ± 6.5 (MCS)	49.4 ± 7.9 (PCS) 51.0 ± 6.9 (MCS)
Baseline EQ-5D	—	0.5 ± 0.3	0.5 ± 0.3
Disc Level (%)			
L2/L3	1 (0.5)	0	1 (1.0)
L3/L4	16 (8.0)	8 (7.9)	8 (8.1)
L4/L5	79 (39.5)	40 (39.6)	39 (39.4)
L5/S1	98 (49.0)	51 (50.5)	47 (47.5)
S1/S2	2 (1.0)	2 (2.0)	0
Other	4 (2.0)	0	4 (4.0)
Operative technique (%)			
Unilateral dissection discectomy	179 (93.7)	92 (92.9)	87 (94.6)
Bilateral muscle dissection and discectomy	12 (6.3)	7 (7.1)	5 (5.4)
Surgeon level (%)			
Consultant	147 (74.6)	72 (72.0)	75 (77.3)
Fellow/registrar	50 (25.4)	28 (28.0)	22 (22.7)

Values represent mean +/− SD or percentage of total. SF-12 corresponds to the short form 12-item survey [physical component score (PCS) and mental component score (MCS), respectively].

and ODI [restricted (51.1 ± 17.8) *vs*. unrestricted (71.1 ± 17.7)] were not significantly different between groups.

At one year after surgery, the primary outcome showed no significant difference between the restricted and unrestricted groups (41.6% *vs*. 36.4%, $P = 0.45$) (Table 2). Analysis of each component of the composite primary outcome measured at one year postoperation demonstrated VAS back improvement ≥ 18 points (61.4% *vs*. 50.5%, $P = 0.12$), VAS leg pain improvement ≥ 25 points (61.4% *vs*. 59.6%, $P = 0.79$), and ODI improvement ≥ 15 points (59.4% *vs* 66.7%, $P = 0.29$), respectively. The need for redo surgery at the index level (2.9% *vs*. 5.5%, $P = 0.68$) similarly did not demonstrate any significant difference between groups.

Secondary endpoints were also similar between groups (Table 3). At one year, reherniation occurred in 7/69 patients (10.1%) in the restricted group compared with 10/71 (14.1%) in the unrestricted group ($P = 0.61$). Both groups demonstrated significant improvements in all functional measures from baseline to 12 months follow-up. Mean ODI scores improved from 46.1% to 17.0% in the restricted arm *versus* 51.1% to 16.8% in the unrestricted arm ($P = 0.06$, between groups). VAS back decreased from 60 to 23.5 points (restricted) and from 62.8 to 24.5 points (unrestricted) ($P = 0.83$). VAS worse leg (leg pain change) decreased from 72.1 to 21.1 in the restricted arm and from 73.8 to 22.8 in the unrestricted arm ($P = 0.67$). In addition, VAS other leg (other leg pain) decreased from 15.1 to 8.8 in the restricted arm and from 16.7 to 10.4 in the unrestricted arm ($P = 0.46$) (Table 4). Mean time for return to work was 12.5 ± 14.3 weeks in the restricted group and 13.2 ± 15.9 weeks in the unrestricted group ($P = 0.83$). EQ-5D index scores improved substantially in both groups, restricted 0.5 ± 0.3 to 0.8 ± 0.2; unrestricted: 0.5 ± 0.3 to 0.8 ± 0.2, with no significant between-group differences [12 mo group comparison, mean difference 0.0 (−0.1 to 0.1), $P = 0.93$]. Mean SF-12 physical component summary (PCS) scores did not significantly change between baseline and one year in both the restricted (48.9–50.3 points) or unrestricted groups (51.0–49.7) [12 mo group comparison, mean difference 0.6 (−1.4 to 2.5) ($P = 0.57$)]. Mental component summary (MCS) mean scores similarly did not significantly change between baseline and one year in both the restricted (50.7 ± 6.7–49.9 ± 6.5) or unrestricted groups (49.4 ± 7.9–50.1 ± 6.0) [12 mo group comparison, mean difference −0.3 (−2.3 to 1.8), $P = 0.81$] (Figure 2).

Electronic activity monitoring demonstrated poor adherence to activity restrictions in the restricted arm, with 2/20 (10%) of the restricted arm patients adhering to sitting restrictions. Sitting duration did not differ significantly between groups (4102 *vs*. 4140 min/wk, $P = 0.89$). A total of 24 and 20 patients for restricted and unrestricted cohorts, respectively, submitted complete activity monitoring data for use in the analysis. Other patient samples were discarded due to insufficient data collection or device loss/failure to return to the hospital.

Most of the patient's time was spent sitting in the first few weeks postoperatively, with standing and walking activity increasing after the first week (Figure 3). There was also no difference in time spent lying, standing, walking, or running between groups (Table 5). No significant differences were seen

TABLE 2. Composite Endpoints Outcomes By Treatment Group

Comparison of Restrictions *Vs.* No Restrictions Groups (Per Protocol Analysis)*

Composite Endpoint Outcomes By Treatment Group*. LOCF, BVR, WVR, and Conservative Analyses for the ITT Cohort Below

Outcome Measure	Cohort	Restrictions Group (%)	No Restrictions Group (%)	Risk Difference (95% CI)	Relative Risk (95% CI)	*P*†	Effect Size‡
Primary outcome							
Composite endpoint success							
Complete	PP	31 (45.6)	32 (47.1)	−1.5% (−18.2 to 15.3)	0.97 (0.67–1.39)	1.000	0.03
LOCF	**ITT**	**42 (41.6)**	**36 (36.4)**	**5.2% (−8.3 to 18.7)**	**1.14 (0.81–1.62)**	**0.449**	**1.07**
BVR	ITT	63 (62.4)	48 (48.5)	13.9% (0.2–27.5)	1.29 (1.00– 1.66)	**0.048**	2.80
WVR	ITT	33 (32.7)	27 (27.3)	5.4% (−7.3 to 18.1)	1.20 (0.78– 1.84)	0.405	1.18
Composite component breakdown							
VAS back pain reduction ≥18							
Complete	PP	47 (69.1)	43 (63.2)	5.9% (−10.0 to 21.8)	1.09 (0.86–1.39)	0.587	0.12
LOCF	ITT	**62 (61.4)**	**50 (50.5)**	**10.9% (−2.8 to 24.6)**	**1.22 (0.95–1.56)**	**0.121**	**2.20**
BVR	ITT	78 (77.2)	61 (61.6)	15.6% (3.0–28.2)	1.25 (1.04–1.51)	**0.017**	3.41
WVR	ITT	47 (46.5)	41 (41.4)	5.1% (−8.6 to 18.9)	1.12 (0.82–1.54)	0.466	1.03
VAS leg pain reduction ≥25							
Complete	PP	49 (72.1)	50 (73.5)	−1.5% (−16.4 to 13.5)	0.98 (0.80–1.20)	1.000	0.03
LOCF	ITT	**62 (61.4)**	**59 (59.65)**	**1.8% (11.8–15.3)**	**1.03 (0.82–1.29)**	**0.796**	**0.37**
BVR	ITT	82 (81.25)	69 (69.7)	11.5% (−0.3 to 23.3)	1.16 (0.99–1.37)	0.059	2.69
WVR	ITT	49 (48.5)	47 (47.5)	1.0 (−12.8 to 14.9)	1.02 (0.77–1.36)	0.883	0.21
ODI improvement ≥15							
Complete	PP	45 (66.2)	55 (80.9)	−14.7% (−29.3 to −0.1)	0.82 (0.67–1.00)	0.080	0.34
LOCF	ITT	**60 (59.4)**	**66 (66.7)**	**−7.3 (−20.6 to 6.1)**	**0.89 (0.72–1.10)**	**0.288**	**−1.51**
BVR	ITT	78 (77.2)	77 (77.8)	−0.6 (−12.1 to 11.0)	0.99 (0.86–1.15)	0.926	−0.13
WVR	ITT	47 (46.5)	56 (56.6)	−10.0% (−23.8 to 3.8)	0.82 (0.63–1.08)	0.156	−2.01
No reherniation							
Complete	PP	62 (91.2)	61 (89.7)	1.5% (−8.4 to 11.4)	1.02 (0.91–1.13)	1.000	0.05
LOCF	ITT	**93 (92.1)**	**88 (88.9)**	**3.2% (−4.9 to 11.3)**	**1.04 (0.95–1.13)**	**0.442**	**1.09**
BVR	ITT	94 (93.1)	89 (89.9)	3.2% (−4.6 to 10.9)	1.04 (0.95–1.13)	0.422	1.14
WVR	ITT	62 (61.4)	61 (61.6)	−0.2% (−13.7 to 13.3)	1.00 (0.80–1.24)	0.973	−0.05
No additional surgery							
Complete	PP	66 (97.1)	64 (94.1)	2.9% (−3.9 to 9.8)	1.03 (0.96–1.11)	0.680	0.15
LOCF	ITT	**99 (98.0)**	**94 (94.9)**	**3.1% (−2.0 to 8.2)**	**1.03 (0.98–1.09)**	**0.277**	**1.71**
BVR	ITT	99 (98.0)	95 (96.0)	2.1% (−2.7 to 6.8)	1.02 (0.97–1.07)	0.442	1.22
WVR	ITT	67 (66.3)	69 (69.7)	−3.4% (−16.3 to 9.6)	0.95 (0.79–1.15)	0.611	−0.72

Bold values refer to LOCF analysis.
Values are n (%) unless otherwise indicated.
Risk difference calculated as restrictions minus no restrictions.
†*P*-value: Fisher exact test for categorical variables.
‡Effect size: Cohen h for proportions; 0.2 = small, 0.5 = medium, 0.8 = large effect, $*P<0.05$, $**\ P<0.01$, $***\ P<0.001$.
Values represent a percentage or a raw numerical value with 95% CI as stated.
*BVR indicates best value replacement; ITT, intention to treat; LOCF, last observation carried forward; PP, per protocol; WVR, worst value replacement.

in secondary outcomes between adherent and nonadherent groups, although this analysis was limited by a small sample size in the adherent group (Table 6). When analysing all patients (regardless of their randomization), by their adherence to the sitting restrictions (< 30 min per 2 h period), there was no significant difference in VAS leg, VAS back, and ODI between those who restricted their sitting and those who did not, although analysis was limited by small sample size (Table 6).

DISCUSSION

This randomized, single-blinded controlled trial demonstrated that assigning specific postoperative activity restrictions in the first month after lumbar microdiscectomy did not affect relief of leg and back pain, quality of life, and reherniation rate. Patients were frequently nonadherent to activity restrictions, and sitting duration did not correlate with recovery.

Recurrent disc herniation significantly impedes recovery, return to work, and reduces quality of life.[4] Risk factors for reherniation include smoking, diabetes, disc protrusion, Modic changes, disc height, segmental instability, and large annular defects.[6,13,14] Surgeons commonly prescribe a period of activity restriction after surgery to reduce load on the disc to prevent reherniation. This often includes avoiding prolonged sitting or heavy lifting.[8,15] The pathophysiological basis of these recommendations appears sound; compared with standing, intradiscal pressure increases 40% when sitting, doubles when bending forward, and increases threefold to fourfold when lifting.[16] Discectomy also significantly increases rotational instability at the operated level,[17] which may predispose to reherniation with these movements.

Current evidence does not support any benefit for postoperative activity restrictions,[18] however, quality data is lacking. Early reintroduction of activity may improve satisfaction with surgery[19] and help the way patients cope with pain.[20] Conversely, imposing mobility restrictions

TABLE 3. Surgical Endpoints: Disc Reherniation and Additional Surgery By Treatment Group

Surgical Endpoints: Disc Reherniation and Additional Surgery Comparison of Restrictions vs. No Restrictions Groups*

	Treatment Groups				Statistical Results	
Outcome Measure	Restrictions Group	No Restrictions Group	Statistical Test	*P*§ (Parametric)†	*P*§ (Nonparametric)†	Effect Size‡
Disc Reherniation	10/101 (9.9%)	13/99 (13.1%)	χ^2 test	—	0.621	−0.101
Reherniation at three months	3/68 (4.4%)	4/79 (5.1%)	χ^2 test	—	1.000	0.031
Reherniation at six months	6/72 (8.3)	7/74 (9.5%)	χ^2 test	—	1.000	0.040
Reherniation at 12 mo	7/69 (10.1%)	10/71 (14.1%)	χ^2 test	—	0.649	0.121
Risk difference (%)	Reference	3.2 (−5.6 to 12.1)	Risk difference	> 0.05	—	0.032
Relative risk	Reference	1.326 (0.61 to 2.883)	Relative risk	—	—	0.282
Odds ratio	Reference	1.373 (0.525 to 3.699)	χ^2 test	—	0.621	0.317
Hazard ratio (Cox model)	Reference	1.354 (0.594 to 3.089)	Cox proportional hazards	0.471	—	0.303
Efficacy (1 - HR)	Reference	−0.354	1-hazard ratio	—	—	0.354
Clinical equivalence	Reference	Not Equivalent (Bounds: 0.8 to 1.2)	Equivalence testing	> 0.05	—	Uncertain
Additional surgery						
Additional surgery at three months	0/68 (0.0%)	4/80 (5.0%)	χ^2 test	—	0.174	0.451
Additional surgery at six months	2/71 (2.8%)	4/74 (5.4%)	χ^2 test	—	0.715	0.132
Additional Surgery at 12 mo	2/69 (2.9%)	4/73 (5.5%)	χ^2 test	—	0.729	0.130

Additional surgery reported as n and the overall percentage.
*Values are n/total (%) for categorical variables.
†Parametric tests: Fisher exact test for categorical variables; nonparametric: χ^2 test.
‡Effect size: Cohen h for categorical variables, log ratio for odds/hazard ratios.
§*$P<0.05$, **$P<0.01$, ***$P<0.001$.
‖Clinical equivalence bounds: HR 0.8–1.2 (20% difference threshold).

may increase fear-avoidance behaviors and increase time off work unnecessarily.[21] Prospective single arm studies suggest that postoperative restrictions delay recovery and return to work without significantly reducing the risk of reherniation or improving pain and functional outcomes.[7] A previous trial assessed the effects of a short (two weeks) versus long (six weeks) duration of restrictions after lumbar microdiscectomy, and found no difference in leg pain, back pain, and reherniation rate.[22] Restrictions in this study included avoiding bending at the waist, lifting more than 10 lbs (~4.5 kg), and twisting; sitting was not specifically limited. Adherence to restrictions was not assessed and outcome assessors were not blinded. Our study randomized patients to either restrictions or no restrictions during the first month after discectomy, with blinded assessment of outcome. We demonstrated that assigning activity restrictions to patients did not affect pain and functional outcomes or the risk of reherniation.

Patients desire clarity about postoperative activity limitations, as they are often fearful of pain, re-injury, and

TABLE 4. Functional Endpoints: Preoperative to One-Year Comparison

Functional Endpoints: Comparison of Outcomes Between Groups

	Restricted Activity		Unrestricted Activity			
Outcome	Restricted Group Mean (SD)	n	Unrestricted Group Mean (SD)	n	*P*	Effect Size
Preoperative back pain (VAS)	60 (28.2)	96	62.8 (24.2)	92	0.833	−0.015
One-year back pain (VAS)	23.5 (24.4)	67	24.5 (22.8)	72	0.626	−0.041
Back pain change (VAS)	−35.6 (38.3)	63	−37.6 (32.2)	66	0.875	−0.014
Preoperative leg pain (VAS)	72.1 (22.8)	96	73.8 (20.1)	92	0.799	−0.019
One-year leg pain (VAS)	21.1 (25.5)	67	22.8 (24.3)	72	0.670	−0.036
Leg pain change (VAS)	−50.5 (36.3)	63	−50.9 (32.7)	66	0.843	−0.017
Preoperative other leg pain (VAS)	15.1 (21.9)	94	16.7 (24.2)	93	0.943	−0.005
One-year other leg pain (VAS)	8.8 (17.9)	67	10.4 (19.7)	72	0.500	−0.057
Other leg pain change (VAS)	−5.3 (25.1)	62	−3 (29.3)	67	0.455	−0.066
Preoperative ODI score (%)	46.1 (17.2)	98	51.1 (17.8)	90	0.114	−0.115
One-year ODI score (%)	17 (15.7)	68	16.8 (14.9)	73	0.919	−0.009
ODI change (Preoperative to one-year)	−25.4 (20.3)	66	−33.1 (21.3)	70	0.058	−0.162

Values reported as mean (SD).
Effect sizes are calculated using Cohen d for parametric tests and r for nonparametric tests. *P*-values <0.05 are considered statistically significant.
ODI indicates Oswestry Disability Index (0%–100%); VAS, Visual Analog Scale (0–100).

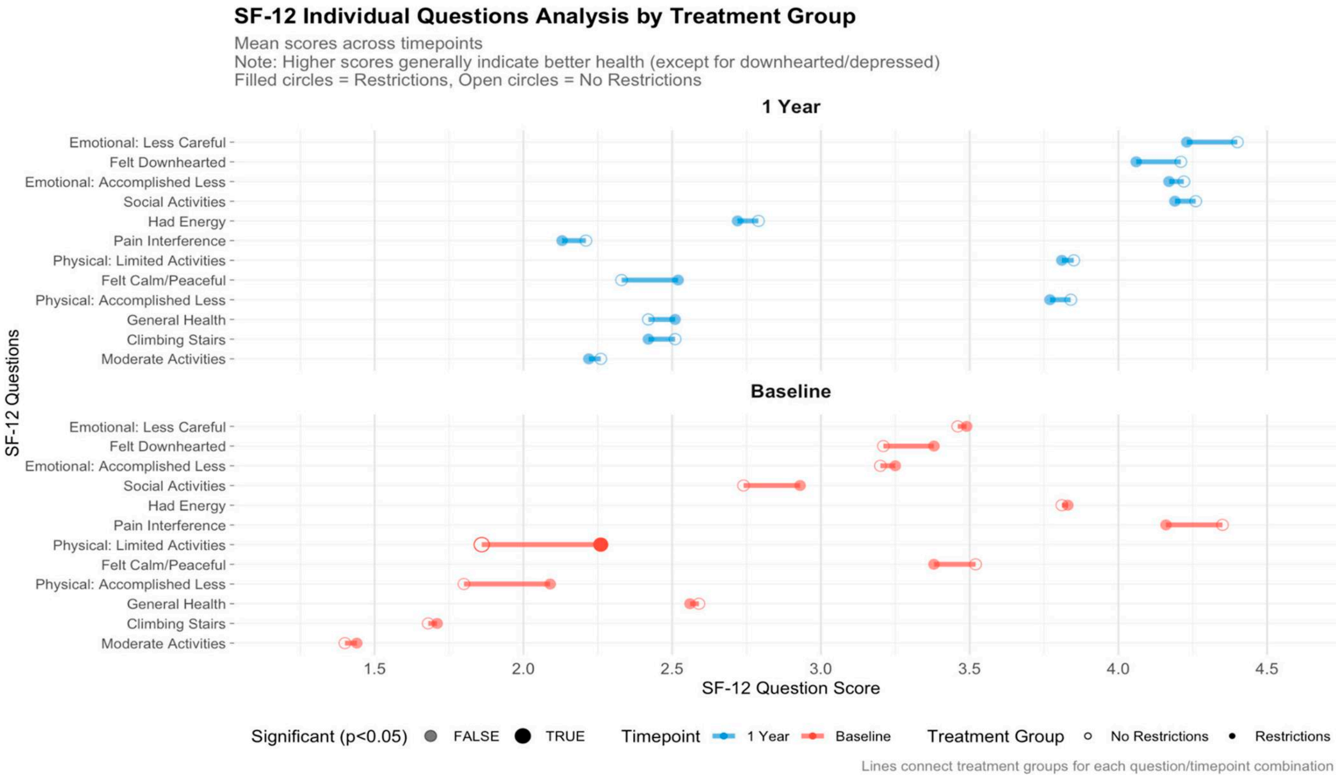

Figure 2. 12-Item short form survey (SF-12) score breakdown compared over time and by treatment group.

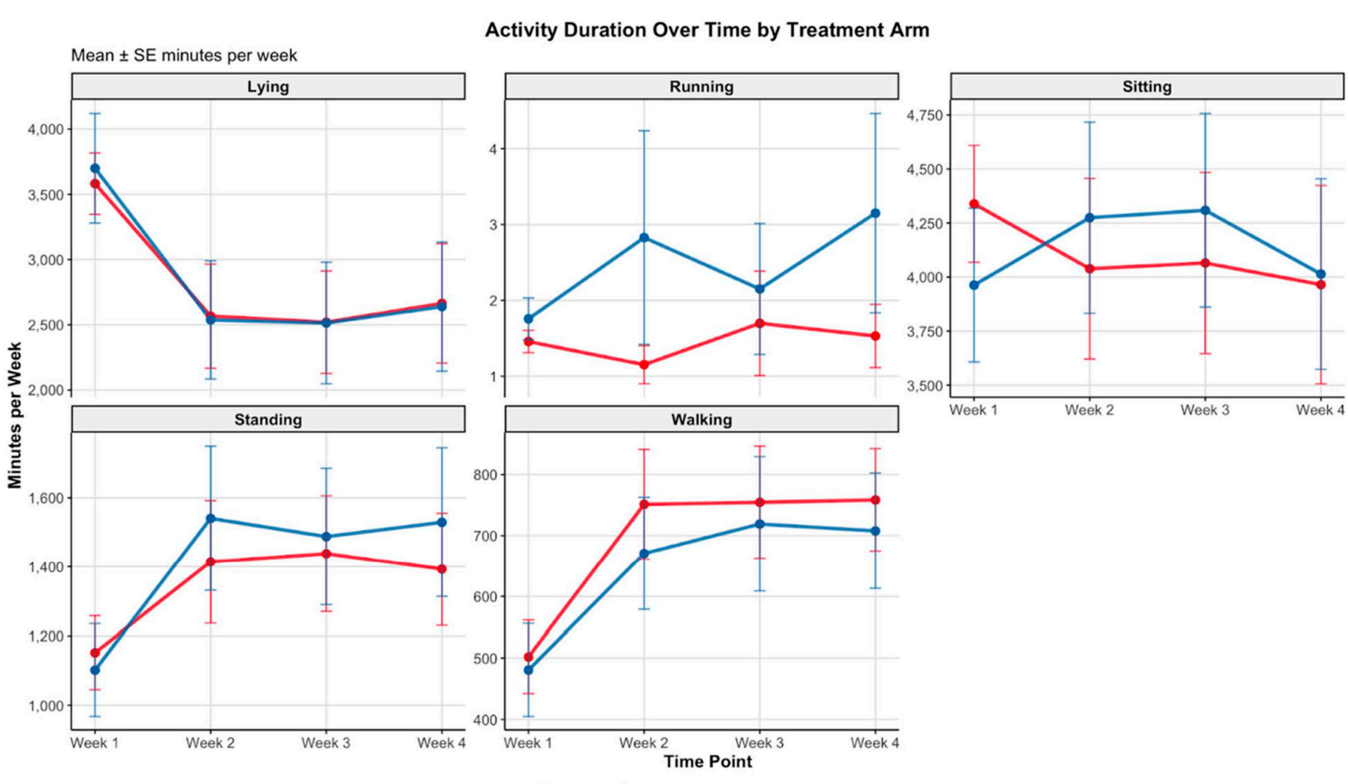

Figure 3. Activity duration between groups: minutes per week (across four weeks).

TABLE 5. Activity Comparison Between Groups: Duration and Intensity Across Four Weeks

Activity Group Comparison (Minutes and METs) Comparison of Restrictions *vs.* No Restrictions Groups

	Treatment Groups		Statistical Results			
Outcome Measure	**Restrictions Group**	**No Restrictions Group**		***P***	**T-statistic**	**Cohen d**
Activity duration (min/wk)	Mean ± SD	Mean ± SD	Difference (95% CI)			
Lying	2832.97 ± 1885.56	2847.73 ± 2076.41	14.76 (−610.23 to 580.72)	0.961	−0.049	0.007
Sitting	4102.29 ± 1863.61	4140.29 ± 1863.61	38.24 (−603.42 to 526.94)	0.8939	−0.134	0.02
Standing	1349.19 ± 761.84	1414.86 ± 861.58	65.67 (−310.16 to 178.82)	0.5965	−0.53	0.081
Walking	691.66 ± 411.53	644.26 ± 423.05	−47.04 (−77.85 to 171.93)	0.4582	0.744	−0.113
Running	1.46 ± 2.07	2.471 ± 4.7	1.012 (−2.14 to 0.11)	0.0769	−1.787	0.288
Activity intensity (METs/wk)						
Lying	0.01 ± 0.02	0.01 ± 0.01	−0.002 (−0.004 to 0.007)	0.5193	0.646	−0.092
Sitting	29.49 ± 10.18	29.01 ± 9.44	−0.48 (−2.44 to 3.41)	0.745	0.326	−0.049
Standing	66.53 ± 28.67	77.52 ± 89.19	10.99 (−31.63 to 9.65)	0.293	−1.058	0.172
Walking	222.73 ± 106.86	308.92 ± 372. 11	86.19 (−171.63 to −0.74)	**0.0481**	−2.004	0.328
Running	11989.48 ± 7684.31	12013.64 ± 9824.2	24.16 (−2691.2 to 2642.88)	0.9857	−0.018	0.003

Values reported as mean ± SD.

"undoing" the effects of surgery.[23] However, postoperative care guidelines vary widely among surgeons. This variability often causes confusion for patients, who may receive conflicting advice from multiple sources, and poses challenges for nursing and allied health staff tasked with patient education. Our results confirm that assigning postoperative activity restrictions does not affect outcome after lumbar microdiscectomy. These results support unrestricted activity for patients after discectomy, providing much needed clarity for patients, reinforcing patient autonomy, potentially reducing fear-avoidance behaviors associated with recovery, and encouraging return to work. An unrestricted approach could enhance patient confidence, outcomes, and satisfaction, ultimately streamlining postoperative care.

Our study demonstrated that when movement is objectively monitored using wearable accelerometers, patients are frequently nonadherent to their activity restrictions. Moreover, most patients adopt a similar level of activity after surgery, regardless of instructions from their surgeon. Patients predominantly engaged in sitting during the first postoperative week, with the proportion of time spent standing and walking increasing as recovery progressed. This is the first randomized trial utilizing wearable accelerometers to assess adherence to postoperative restrictions. The lack of adherence may explain the lacking benefit in prescribing activity restrictions observed in prior studies and highlights the challenge of enforcing strict protocols to patients in their early recovery period. Adherence issues may also further explain the lower than expected composite success rates produced from the current study as compared with existing literature. In addition, higher than expected patient drop-out likely contributed, wherein lost patients may disproportionately represent those recovering very well, to no longer seek ongoing intervention, and those doing very poorly, to become disengaged from ongoing study.

Limitations

The study has several limitations. The sample size, while adequate for primary outcomes, was not sufficiently powered to detect a difference in uncommon outcomes, such as reherniation. There was also a relatively high drop-out rate, 31% and 28%, respectively, we accounted for a 30% drop-out rate. The follow-up period (12 mo) may be insufficient to

TABLE 6. Adherence Comparison for All Patients: Secondary Outcomes

Activity Group Comparison (VAS and ODI) Comparison of Adherent *vs.* Nonadherent Patients

Outcome	**Adherent Group Mean (SD)**	**n**	**Nonadherent Group Mean (SD)**	**n**	***P***	**Effect Size**
Preoperative back pain (VAS)	58.6 (15.9)	5	41.1 (21.5)	39	0.100	0.248
One-year back pain (VAS)	25 (22.1)	3	27 (27.2)	35	0.871	0.026
Back pain change (preoperative to one-year)	−24.7 (33.6)	3	−14.1 (30.2)	35	0.705	0.061
Preoperative leg pain (VAS)	26.4 (20.2)	5	18.6 (20.6)	39	0.255	0.172
One-year leg pain (VAS)	41.7 (27.5)	3	23.1 (27.1)	35	0.254	0.185
Leg pain change (preoperative to one-year)	27.3 (34.8)	3	5.7 (31.4)	35	0.386	0.141
Preoperative other leg pain (VAS)	2.6 (3.7)	5	2 (5.2)	39	0.348	0.141
One-year other leg pain (VAS)	3.3 (4.9)	3	10.1 (17.2)	35	0.512	0.106
Other leg pain change (preoperative to one-year)	2 (4.6)	3	7.9 (18.6)	35	0.870	0.026
One-month ODI score (%)	33 (18.9)	4	23.1 (15.3)	39	0.336	0.147
One-year ODI score (%)	35.3 (12.9)	3	18.4 (16.3)	36	0.086	0.275

Adherent patients followed activity restrictions regardless of original treatment assignment. Effect sizes calculated using r for Mann-Whitney *U* tests. *P*-values < 0.05 are considered statistically significant.

ODI indicates Oswestry disability index (0%–100%); VAS, visual analog scale (0–100).

capture long-term differences in outcome. Patient adherence to assigned protocols was poor, and thus, the effect of strict adherence on outcome is unknown. The study population was limited to single-level unilateral lumbar disc herniation, limiting generalizability to other spine pathology. In addition, primary composite endpoint success irrespective of group was lower than expected. It was anticipated that a success rate of between 70% and 90% would be attained, as evidenced in existing literature.[24] Each individual threshold (18-point VAS back reduction, 25-point VAS leg reduction, 15-point ODI improvement, no reherniation, no secondary intervention) must be met simultaneously, and this conjunction likely proved far more stringent than anticipated. The present study does not retain statistical power to detect significant differences between composite success rates noted as a ∼1.5% difference between arms versus the 20% original detection ability; results should be interpreted with some caution.

Finally, activity assessment with wearable accelerometers, while a novelty and strength of our study, may not fully capture the nuances of patient activity or comfort levels.

CONCLUSION

This randomized, blinded trial suggests that assigning specific postoperative activity restrictions did not affect outcomes after lumbar microdiscectomy. Patients were often noncompliant with restrictions, and activity, prolonged sitting in particular, did not jeopardize recovery. Within the limitations of this study, it would suggest that recommending patient activity as tolerated after surgery is safe. This also potentially encourages faster recovery, earlier return to work, and greater satisfaction with surgery.

➢ Key Points

- ❑ Primary outcome: At one year postsurgery, there was no significant difference between restricted and unrestricted groups in the composite primary endpoint (41.6% *vs.* 36.4%, $P=0.45$), including VAS back pain, VAS leg pain, and ODI improvements.
- ❑ Individual measures: Both groups showed significant improvements in all functional measures from baseline to 12 months, with VAS back pain decreasing from around 60 to 23 to 25 points and ODI scores improving from 46%–51% to 16%–17%.
- ❑ Reherniation rates: The need for repeat surgery at the index level was similarly low in both groups (2.9% *vs.* 5.5%, $P=0.68$), showing no significant difference between restricted and unrestricted postoperative protocols.

References

1. Bailey CS, Rasoulinejad P, Taylor D, et al. Surgery versus conservative care for persistent sciatica lasting 4 to 12 months. *N Engl J Med*. 2020;382:1093–1102.
2. Peul WC, van Houwelingen HC, van den Hout WB, et al. Surgery versus prolonged conservative treatment for sciatica. *N Engl J Med*. 2007;356:2245–2256.
3. Weinstein JN, Tosteson TD, Lurie JD, et al. Surgical vs nonoperative treatment for lumbar disk herniation: the Spine Patient Outcomes Research Trial (SPORT): a randomized trial. *JAMA*. 2006;296:2441–2450.
4. Shepard N, Cho W. Recurrent lumbar disc herniation: a review. *Global Spine J*. 2019;9:202–209.
5. Billy GG, Lemieux SK, Chow MX. Changes in lumbar disk morphology associated with prolonged sitting assessed by magnetic resonance imaging. *PM R*. 2014;6:790–795.
6. Zileli M, Oertel J, Sharif S, Zygourakis C. Lumbar disc herniation: prevention and treatment of recurrence: WFNS spine committee recommendations. *World Neurosurg X*. 2024;22:100275.
7. Carragee EJ, Han MY, Yang B, Kim DH, Kraemer H, Billys J. Activity restrictions after posterior lumbar discectomy. A prospective study of outcomes in 152 cases with no postoperative restrictions. *Spine (Phila Pa 1976)*. 1999;24:2346–2351.
8. Daly CD, Lim KZ, Ghosh P, Goldschlager T. Perioperative care for lumbar microdiscectomy: a survey of Australasian neurosurgeons. *J Spine Surg*. 2018;4:1–8.
9. Delgado DA, Lambert BS, Boutris N, et al. Validation of digital visual analog scale pain scoring with a traditional paper-based visual analog scale in adults. *J Am Acad Orthop Surg Glob Res Rev*. 2018;2:e088.
10. Fairbank JC, Couper J, Davies JB, O'Brien JP. The Oswestry low back pain disability questionnaire. *Physiotherapy*. 1980;66:271–273.
11. Ware J Jr, Kosinski M, Keller SD. A 12-item short-form health survey: construction of scales and preliminary tests of reliability and validity. *Med Care*. 1996;34:220–233.
12. EuroQol–a new facility for the measurement of health-related quality of life. *Health Policy*. 1990;16:199–208.
13. Huang W, Han Z, Liu J, Yu L, Yu X. Risk factors for recurrent lumbar disc herniation: a systematic review and meta-analysis. *Medicine (Baltimore)*. 2016;95:e2378.
14. Miller LE, McGirt MJ, Garfin SR, Bono CM. Association of annular defect width after lumbar discectomy with risk of symptom recurrence and reoperation: systematic review and meta-analysis of comparative studies. *Spine (Phila Pa 1976)*. 2018;43:E308–E315.
15. Magnusson ML, Pope MH, Wilder DG, Szpalski M, Spratt K. Is there a rational basis for post-surgical lifting restrictions? 1. Current understanding. *Eur Spine J*. 1999;8:170–178.
16. Nachemson AL. Disc pressure measurements. *Spine (Phila Pa 1976)*. 1981;6:93–97.
17. Goel VK, Goyal S, Clark C, Nishiyama K, Nye T. Kinematics of the whole lumbar spine. Effect of discectomy. *Spine (Phila Pa 1976)*. 1985;10:543–554.
18. McGregor AH, Burton AK, Sell P, Waddell G. The development of an evidence-based patient booklet for patients undergoing lumbar discectomy and un-instrumented decompression. *Eur Spine J*. 2007;16:339–346.
19. Kjellby-Wendt G, Styf J. Early active training after lumbar discectomy. A prospective, randomized, and controlled study. *Spine (Phila Pa 1976)*. 1998;23:2345–2351.
20. Kjellby-Wendt G, Styf J, Carlsson SG. Early active rehabilitation after surgery for lumbar disc herniation: a prospective, randomized study of psychometric assessment in 50 patients. *Acta Orthop Scand*. 2001;72:518–524.
21. Claus D, Coudeyre E, Chazal J, Irthum B, Mulliez A, Givron P. An evidence-based information booklet helps reduce fear-avoidance beliefs after first-time discectomy for disc prolapse. *Ann Phys Rehabil Med*. 2017;60:68–73.
22. Bono CM, Leonard DA, Cha TD, et al. The effect of short (2-weeks) versus long (6-weeks) post-operative restrictions following lumbar discectomy: a prospective randomized control trial. *Eur Spine J*. 2017;26:905–912.
23. Williamson J, Bulley C, Coutts F. What do patients feel they can do following lumbar microdiscectomy? A qualitative study. *Disabil Rehabil*. 2008;30:1367–1373.
24. Weinstein JN, Lurie JD, Tosteson TD, et al. Surgical vs nonoperative treatment for lumbar disk herniation: the Spine Patient Outcomes Research Trial (SPORT) observational cohort. *JAMA*. 2006;296:2451–2459.